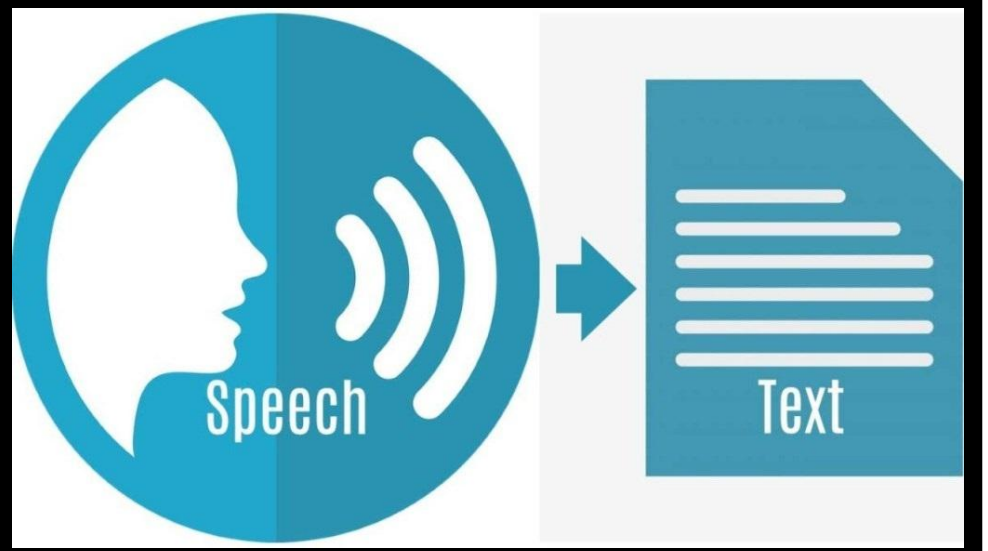
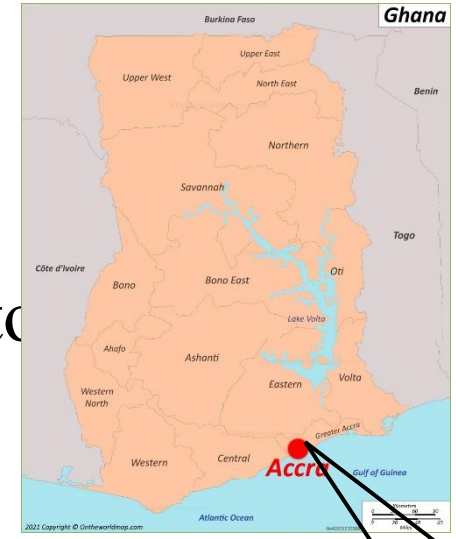


Building a High-Accuracy Ga ASR system using the transformer-based Whisper Architecture

Presentation made by Henry Dakpo

Why Ga ASR?

- Language inequality in speech to text technologies such as Siri, Google Voice etc.
- Ghana is multilingual, but ASR tools only understand English
- The native language of Ghana's capital, Accra, is Ga (Ga-Adangbe)
- Having a Ga-ASR model improves communication among humans, support human-computer interaction and helps preserve the Ga language and culture



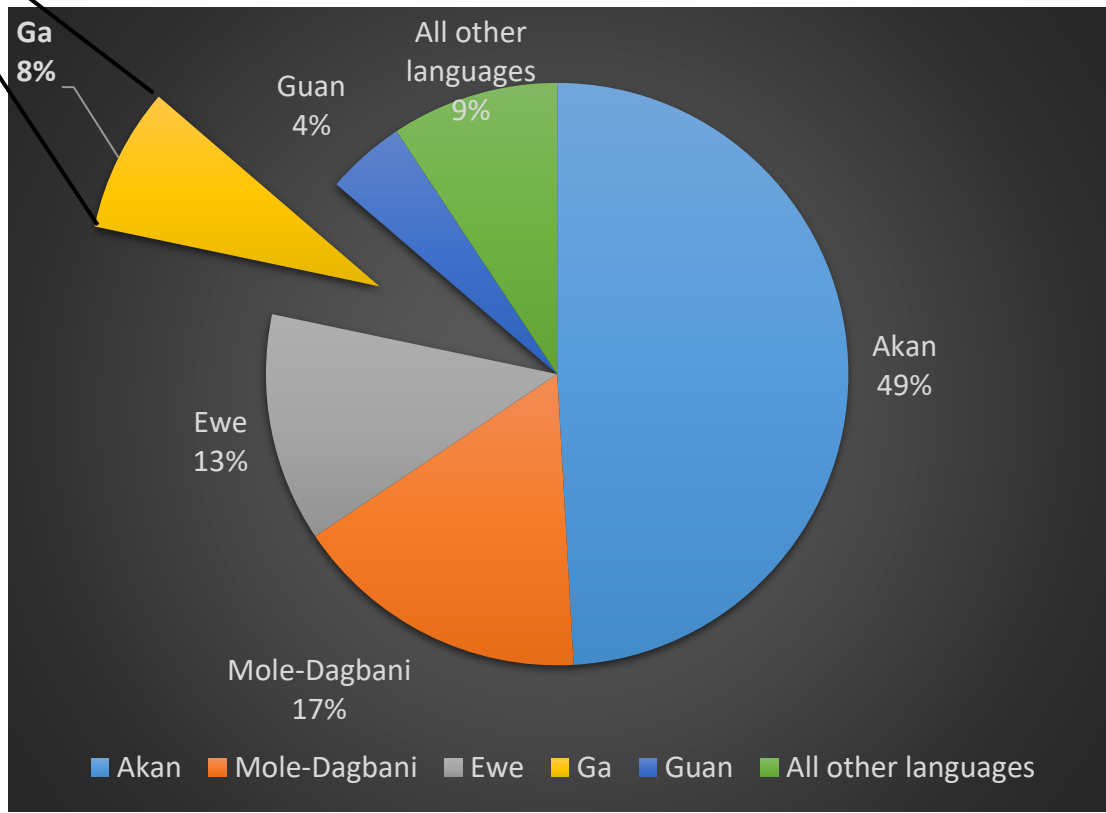


Figure modified from Ansah (2014)

Goal and objectives

Goal:

- To develop and optimize an ASR system for Ga language using Whisper transformer architecture, with the long-term aim of achieving high transcription accuracy for a low-resource language

- Prepare and preprocess Ga speech–text dataset
- Train and evaluate an initial Whisper-based ASR model
- Build a working prototype that converts Ga audio text

Objective 1 (Baseline System - Completed)



- Increase training data & improve data quality
- Optimize training (hyperparameters, more epochs, unfreeze encoder)
- Enhance text normalization and preprocessing
- Explore larger Whisper models for better WER

Objective 2 (Improve Accuracy - Ongoing)



Why did I choose Whisper over other ASR Models?

- Pre-trained on a large multilingual speech dataset
- Works well even with small datasets
- Handles accents and background noise better than many models
- Easy to fine-tune
- Uses Modern transformer architecture



Methodology to Achieve Objective 1

- Set up required tools (Colab Pro, Whisper model, Python libraries, Artificial Intelligence, etc.)
- Downloaded the Ga-Adangbe dataset from Lacuna Fund
- Uploaded dataset to drive and loaded it into Colab



Methodology to Achieve Objective 1

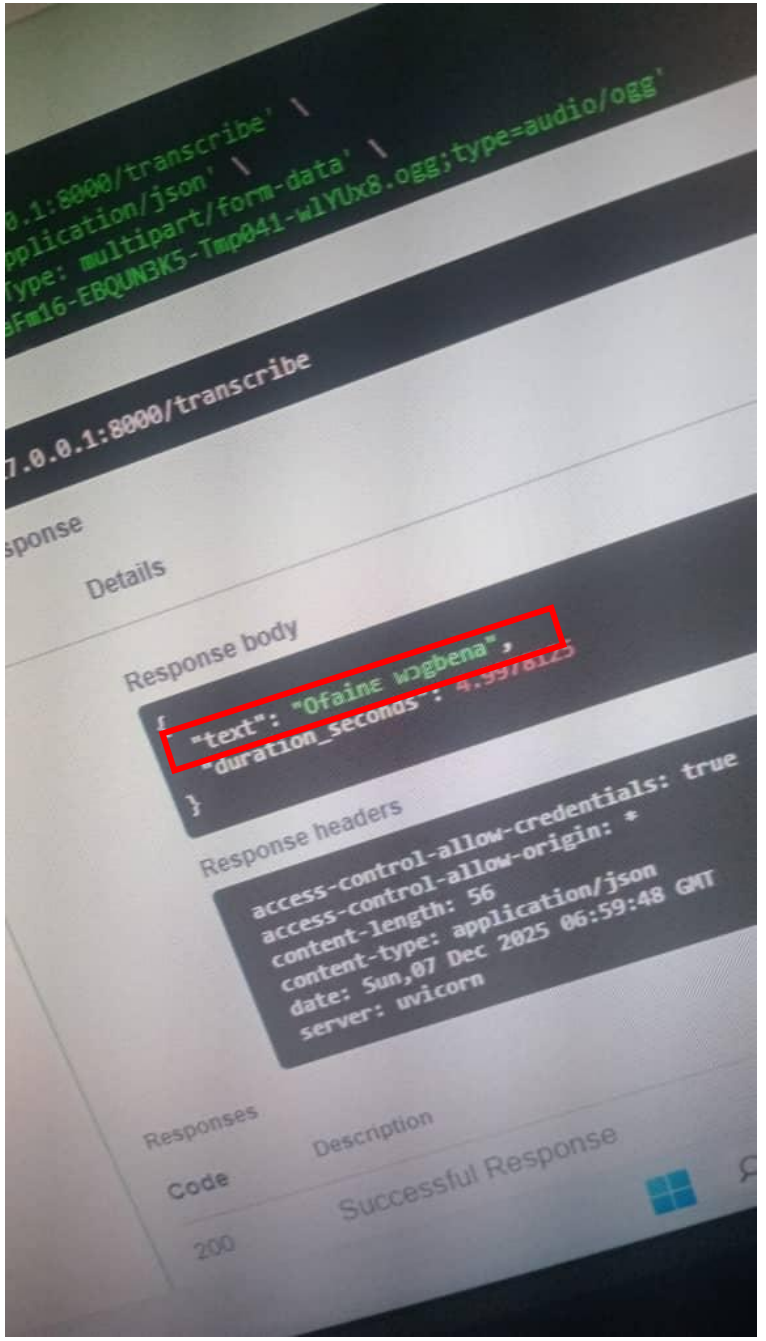
- Analyzed the data for accuracy, validity and consistency
- Preprocessed the dataset (audio normalization & text cleaning)
- Trained the machine to listen, understand and transcribe



Results for Objective 1

- **Positive Outcomes (Strengths)**
  - Training Dynamics Look Good
  - Training time for quality output is faster compared to other tools
  - Whisper can be fine-tuned on Ga





Results for Objective 1

- **Negative Outcomes (Weakness)**
  - Errors in transcripts
  - Weights are closed



```
[ ]
from google.colab import files

print("Upload a Ga .wav file for testing")
uploaded = files.upload()

for fname in uploaded.keys():
    print(f"\nTranscribing: {fname}")
    text = transcribe_file_ga(fname)
    print("Transcription:", text)
```

```
Upload a Ga .wav file for testing
Choose Files No file chosen    Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.
Saving GaFm16-EBQUN3K5-Tmp052-8ibyvw.ogg to GaFm16-EBQUN3K5-Tmp052-8ibyvw.ogg

Transcribing: GaFm16-EBQUN3K5-Tmp052-8ibyvw.ogg
Transcription: ofaɔi nɛmi ni jiɔ akɛŋtsu nɔŋmɔɔ.
```

Future Direction/Plan for Objective 2

- **Increase training data**
  - Expanding number of audio samples
  - Using more of the HuggingFace dataset (from 20k to 60k, then to 90k)
  - Augmenting data (noise addition, speed variation)
- **Train for more epochs**
  - From 2 epochs to 4-10 epochs
  - Early stopping to avoid overfitting



Future Direction/Plan for Objective 2

- **Fine-tune more parts of the model**
  - Unfreeze Whisper encoder
  - Fine-tune full transformer instead of decoder-only
- **Hyperparameter tuning**
  - Adjust learning rate
  - Gradient accumulation
  - Batch size
  - Warmup steps



Future Direction/Plan for Objective 2

- **Model scaling**
  - Move from whisper-small to whisper-medium/large if resources allow
  - Larger models often yield lower errors



Thank you

References

- Ansah, G. N. (2014). Re-examining the fluctuations in language in-education policies in post-independence Ghana. *Multilingual education*, *4*(1), 12